

Car Rental Use Case

Our Team

PROJECT OVERVIEW

A structured data workflow for integrating and refining multi-city rental data, encompassing reservations, telematics, maintenance logs, and payment systems to enable reliable analytics and decision-making.



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Problem Statement

Core data challenges discovered during exploration reveal structural inconsistencies, free-text errors, and mixed units. These issues directly affect downstream analytics

Data Inconsistency

Vehicle_ID variations, messy timestamps, mixed odometer/fuel formats. Structural inconsistencies prevent accurate data joining. example includes "45,000 km" vs "45000", minutes like 10:75 and non-standardized IDs

Operational Impact

Pricing inaccuracies, maintenance misses, availability misreports. Dirty data leads to revenue leakage through incorrect billing, missed maintenance windows causing breakdowns, and poor customer experience due to vehicle availability.

Business Risk

Fraud exposure and compliance risks from bad PII handling. Free-text notes containing unmasked PII create GDPR/privacy violations. Incorrect tariff application leads to billing disputes, tax mismatches, and potential regulatory fines.



Dataset overview

Segment of Raw Data

Vehicle_ID	Pickup_TS	Odo_Start	Rate	City
CAR-188	2025-01-22T00:38	30434 km	\$20/day	MUM
CAR_029	2025/02/04 13:33	24285 km	\$20/day	Delhi
CAR_006	2025-01-10T23:38	79,142	Rs1500/day	chennai
CAR411	02/01/2025 13:52	69,349	Rs1500/day	MUM
CAR-327	01/02/2025 02:35	57,384	₹1,800/day	delhi
car-295	04/06/2025 16:29	32076	1800 perday	delhi
car342	2025-06-23 10:25	26291 km	NaN	Mumbai

Segment of Cleaned Data

Customer_ID	Pickup_TS	Odo_Start	Rate	City	Payment
CUST-1176	2025-03-08 03:38:00	51887.0	1840.00	Delhi	UPI
CUST-0561	2025-03-28 17:23:00	43960.0	1800.00	Delhi	CARD
CUST-1389	2025-04-27 11:23:00	37826.0	2500.00	Delhi	CARD
CUST-0580	2025-05-17 02:58:00	60155.0	2300.00	Chennai	UPI
CUST-0298	2025-06-06 19:17:00	24553.0	1835.27	Delhi	CARD
CUST-1129	2025-06-11 20:44:00	35500.0	1500.00	Chennai	CARD

PROVENANCE & SIZE

~5000

Total Rows

~21

Total Columns

~473

Duplicated Rows
(Intentional Noise)

Data Characteristics

- Contains missing and inconsistent values
- Includes duplicate and overlapping bookings
- Mixed timestamp formats (real-world issue)
- Multi-city and multi-vehicle data
- Designed to test data cleaning & validation logic

Major Issues

Inconsistent Vehicle IDs

Invalid Rental Duration

Odometer Unit Inconsistency

Fuel Level Format Variations

Duplicate Reservation Records

*Invalid & Mixed Timestamp
Formats*

Rental Rate Format Differences

Mileage Validation Issues



Tools & Libraries Used



Core programming language used to build the ETL data cleaning process, implement transformation logic, and process large CSV datasets efficiently.



Supports high-performance numerical computations and vectorized operations for efficient data processing.



Used for data ingestion, manipulation, and transformation including duplicate removal, timestamp normalization, unit standardization, and dataset restructuring.



Interactive environment used for developing, testing, and documenting the data cleaning



Used to create visualizations and exploratory analysis plots to understand patterns, anomalies, and data quality issues.



Provides advanced statistical visualizations to analyze relationships, distributions, and trends within the rental dataset.



Used for location-based processing and city normalization, enabling geographic data validation and enrichment.



Used for automated testing of cleaning and transformation steps to ensure reliability and correctness of the process.

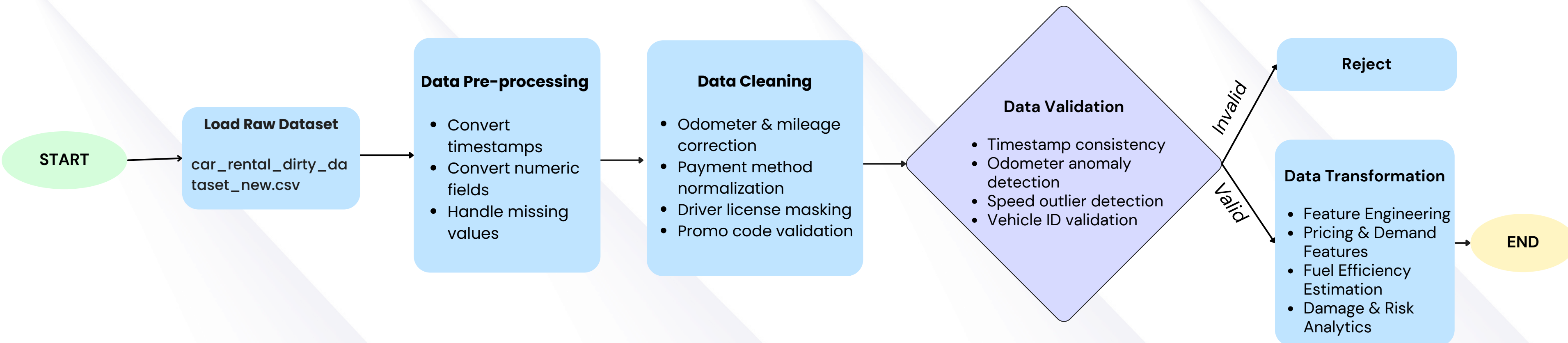
What We Implemented

We built an end-to-end Car Rental Data Cleaning & Transformation workflow in Python to process raw CSV rental data into a clean, analysis-ready dataset.

Additional standardization was applied to ensure uniform formats for Vehicle IDs and city names, along with normalization of fuel levels for accurate analysis.

Data validation ensures consistency and reliability by identifying issues before further analysis.

Key checks: Timestamp consistency, odometer unit standardization, overlap detection, and fuel level validation.



Thank You

We appreciate your time and attention.
We are now ready for your questions and technical discussion.

Car Rental Use Case Team



Key Outcomes

Early validation logic is critical to prevent data pollution. Enforcing canonical naming conventions, standardizing measurement units (km vs miles), and automating test suites ensures scalability and data integrity across multi-city operations.